



Transformation of Hierarchical Multiply Controlled Verbal Relations in Children With Autism in a Game of I Spy

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Abstract

We evaluated the development of mutually entailed arbitrary hierarchical relations and associated transformations of stimulus function across 3 children with autism in a game of I Spy. Top-down hierarchical relational training was efficacious in establishing 4 relational categories (A) containing a total of 5 stimuli (B), where 3 of the stimuli were contained in 2 different categories. Following relational training, all participants demonstrated a transformation of function by identifying the stimuli when provided a multiple verbal stimulus with two category names during I Spy. The procedures were adapted from the PEAK Relational Training System.

Keywords Relational frame theory · Verbal behavior · Transformation of function · Autism

A series of recent studies have begun to establish that common children's games may provide early opportunities to establish complex verbal behavior repertoires and that these activities may be applied in the context of language training for people with autism (Dixon, Blevins, Belisle, & Bethel, 2019; Dixon, McCord, & Belisle, 2018; Dixon, Speelman, Rowsey, & Belisle, 2016). These approaches have incorporated Skinnerian verbal operant theory (Skinner, 1957) with advances in relational frame theory (RFT; Hayes, Barnes-Holmes, & Roche, 2001) to provide a comprehensive account of language learning. Dixon, Blevins, et al. (2019) used a direct reinforcement approach to bring extended verbal utterances under the control of an audience in the context of Show-and-Tell. In another study, Dixon, McCord, and Belisle (2018) conditioned solving word problems as a higher order generalized operant behavior using multiple-exemplar training, after which one child with autism and a typically developing child could solve novel word scrambles in the absence of direct reinforcement. Finally, Dixon et al. (2016) demonstrated greater complexity by establishing coordinated

networks containing known body parts and their synonyms and by demonstrating a transformation of stimulus function (i.e., behaving differently in an untrained task following relational training) in a game of Twister using the transitively related synonyms. A similar theme across these studies was that the procedures were adapted from the PEAK Relational Training System (Dixon, 2016) as part of a comprehensive language training curriculum for use with individuals with autism.

Another common children's game that likely emerges through both verbal operant and relational processes is I Spy. The game involves a person uttering, "I spy with my little eye something that is *X* and *Y*," where another individual is tasked with identifying a stimulus in the environment that contains the properties *X* and *Y*. For example, if *X* was "red" and *Y* was "has wheels," a correct response could involve a red toy truck. Skinner (1957) described this as responding in terms of a multiple stimulus, where both "red" and "wheels" could apply to any number of objects in isolation but the combined stimulus significantly reduces potential correct responses. RFT adds greater depth to this interpretation in terms of hierarchical relational framing, in which classes of stimuli are related into higher order classes, specifying that stimuli are "a member of," "an attribute of," "contained by," or "belong to" other stimuli within those classes (Hayes et al., 2001). Hierarchical relational frames are distinct as a result of three primary characteristics: transitive class containment, asymmetrical class containment, and unilateral property induction.

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As a result, when responding hierarchically, an individual would respond to a stimulus in one class as a member of additional higher order classes, but this relationship is “one-directional,” as the higher order stimuli are not contained within the stimulus, and the specific features of lower order stimuli do not necessarily transfer to the higher order stimuli (for a more in-depth discussion of the characteristics of hierarchical frames, see Slattery & Stewart, 2014).

Returning to the example of I Spy, through a history of verbal relational conditioning, objects are contained within higher order categorical classes. For example, “apple,” “Elmo,” and “red toy truck” are contained in the class “red,” and “blue car” and “red toy truck” are contained in the class “objects with wheels.” Objects within the class are not coordinated, as “apple” is different from “Elmo,” although they are coordinated among one physical dimension (i.e., “redness”). The class cannot be described in terms of stimulus generalization, as the verbal utterance “apple” does not have formal similarity to the verbal utterance “Elmo.” The class also cannot be described in terms of stimulus equivalence or coordination because, although both “red” and “objects with wheels” are entailed with (i.e., related to) “red toy truck,” “red” and “objects with wheels” are not equivalent verbal constructs. The distinction between coordinated and hierarchical networks at this most basic level is shown in Fig. 1. “Red toy truck” (e.g., C in the figure) is the only object that is contained in both the “red” (e.g., A1 in the figure) and “objects with wheels” (e.g., A2 in the figure) hierarchical classes. Another object such as “blue sports car” (e.g., D in the figure), though an object with wheels (contained within A2), is not “red” (not A1) and is therefore not coordinated with the red

toy truck (C) despite mutual participation in the same A2 class. Instead, the blue sports car (D) may be the only object contained in both the A2 class and the “fast vehicles” (A3) class. Therefore, responding to the multiple stimulus in the context of I Spy requires both multiple stimulus control and the transformation of stimulus function in terms of established hierarchical relational frames. If the relations were instead coordinated as shown in the top of the figure, then not only would “red toy truck” and “blue sports car” be treated as the same or equal, but also the categories “red,” “objects with wheels,” and “fast vehicles” would also be considered the same or equal. Sorting of the “blue sports car” in terms of A2 and A3 and “red toy truck” in terms of A1 and A2 would suggest that coordination among all class members did not occur and instead is indicative of the hierarchical relating required to meaningfully engage in the game of I Spy.

Researchers have examined categorical responding with training approaches other than hierarchical relational responding or RFT interpretations in general. For example, children with disabilities have been taught to respond to stimuli categorically with training procedures involving intraverbals, naming, and stimulus equivalence (e.g., Kisamore, Carr, & LeBlanc, 2011; Miguel, Petursdottir, Carr, & Michael, 2008; Dixon et al., 2017). An RFT interpretation including hierarchical responding may be an avenue for understanding additional complexities of categorization, such as the unidirectional formation of categories and coordination of noncoordinated stimuli within a hierarchical class. Although empirical research related to RFT has increased significantly in the past decade as noted in a recent RFT citation analysis, examinations of hierarchical responding have been

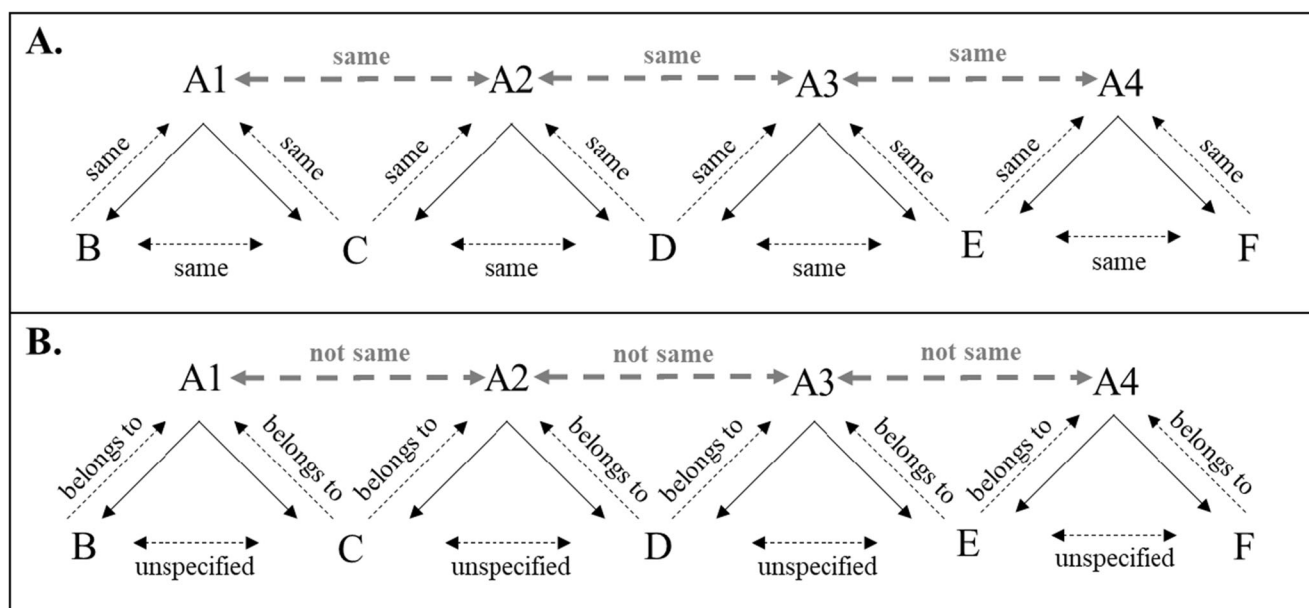


Fig. 1 The distinction between entailed coordinated relations (top: A) and hierarchical relations (bottom: B). Solid black arrows show trained relations, and dashed black lines show potential derived relations within

A categories. Thick, dashed gray lines show the derivations between A categories (i.e., coordination = same; hierarchy = not same)

more limited, with only 10 of the 200 empirical RFT articles included addressing hierarchical frames specifically (O'Connor, Farrell, Munnely, & McHugh, 2017). Additional research examining hierarchical responding in both basic and applied experimental arrangements is needed.

In the present study, we sought to evaluate whether conditioning of early arbitrary hierarchical relations, in which verbal stimuli participated in multiple relational classes, would produce a transformation of stimulus function in I Spy. The target relational classes were considered an early form of hierarchical relating as the relations were mutually entailed, whereas a more complex and complete demonstration requires combinatorial entailment. This approach extends prior research suggesting that common children's games may provide verbal relational training opportunities with the potential for application with children with autism. As in the prior studies, procedures were adapted from PEAK, specifically the Transformation module, to aid in future replication both clinically and in research.

Method

Participants, Settings, and Materials

Participants were three males diagnosed with autism spectrum disorder. Joseph was a 13-year-old who attended a Midwestern school specialized for children with autism. William, 9-years-old, and Daniel, 8-years-old, attended a university applied behavior analysis clinic for children with language and cognitive deficits. All three participants had advanced language capabilities and were able to communicate vocally with school and clinic staff.

Prior to the study, researchers completed PEAK Transformation module preassessments' (PEAK-T-PA) expressive and receptive subtests (PEAK-T-PA-E/R) for all participants. The PEAK-T-PA assesses an individual's relational responding repertoire across six relational frame families: coordination, comparison, opposition, distinction, hierarchy, and deictic. The expressive and receptive tests increase in complexity within each frame, beginning with nonarbitrary stimuli, proceeding to culturally established stimuli, or those that are commonly defined by the verbal community, followed by arbitrary stimuli. For the purpose of this study, combined receptive and expressive hierarchy subtest scores (each consisted of 32 total possible points) were considered when selecting participants (Richard: 16 of 32; Liam: 24 of 32; Phillip: 24 of 32). All three participants' scores on the hierarchy subtests indicated that nonarbitrary and culturally established hierarchical relations were established skills, and thus the participants were predicted to be capable of learning arbitrarily applicable relations. IQ scores were also accessed from the participants' files prior to the study and indicated composite IQ scores of 74 for Joseph, 92 for William, and 112 for Daniel.

Joseph's sessions were conducted in a classroom with a desk, two chairs, and the stimuli necessary for the experiment. The classroom used was separate from other students. William's sessions were conducted in a small therapy room with a table, two chairs, toys and art materials, and the stimuli necessary for the experiment. Daniel's sessions were conducted in a small therapy room with a table, two chairs, and materials for the experiment. The training procedures and materials were adapted from the PEAK-T program 13I—I Spy Something Arbitrary, which includes transportation images and arbitrary words. Transportation stimuli included pictures of a bird, airplane, helicopter, semitruck, and car. Arbitrary word stimuli were presented to participants vocally but were printed on a piece of paper for participants to refer to when responding. The four word stimuli used were *SAN*, *HEF*, *FRA*, and *WOZ*.

All participants received verbal praise as reinforcement for correct responding during the study and followed individual token systems already in place from their current teachers or therapists. Participants were able to exchange tokens for access to an iPad or a break from the workspace.

Procedure

A multiple-baseline design across participants was used for the current study. Phases included baseline test probes for two sets of relations (top-down hierarchical relations and bottom-up hierarchical relations), as well as the transformation of stimulus function in a novel task, top-down relational training, and follow-up test probes. For Joseph and William, probes conducted following training trials tested first for the emergence of mutually entailed relations, then for the transformation of stimulus function within a game of I Spy. For Daniel, the transformation test was conducted before the tests of mutual entailment. This change was made to control for sequencing effects. The dependent variable for the study was the percentage of correct responding for training trials and test trials. The independent variable was the implementation of procedures adapted from the selected PEAK-T program. Responding was recorded as the percentage of correct responding within an eight-trial block for all stimulus relations and a six-trial block for the transformation I Spy game. Interobserver agreement was calculated for 53.62% of trials using the formula for total agreement ($[\text{total agreements} / \text{total disagreements}] \times 100$); 100% agreement was reported across trials.

Baseline Baseline included trial blocks from the top-down training, mutual entailment tests, and transformation tests. The stimulus presentation order was random for each trial. During the baseline phase, scores were recorded as correct or incorrect and no form of prompting, feedback, or reinforcement for responding was provided. Participants were given

short breaks between trial blocks. Baseline trial blocks were staggered across participants. Following the baseline phase, training and testing phases were conducted where top-down relations were trained and bottom-up relations were tested. In addition, a test for the transformation of stimulus function in a novel context was conducted again following training.

Top-down training In this phase, participants were taught to match two types of transportation (B and C or B and D) that belonged to a single category when provided an arbitrary word (A; A-B/C or A-B/D). The stimuli A-B/C/D hierarchy used in the present study are presented in Fig. 2. Participants were presented with all five picture cards in a random array on the table. Participants were then asked, “Which two belong to [SAN/HEF/FRA/WOZ].” For example, when asked, “Which two belong to SAN?” (A), the participant was taught to select the pictures of the airplane (B) and helicopter (C). When asked, “Which two belong to HEF?” (A), the participants were taught to select the pictures of the airplane (B) and bird (D). If the participant selected the correct two picture stimuli, he was provided with vocal praise. If the participant responded incorrectly, he received a verbal or visual prompt to select the correct picture cards. If further prompting was required, the researchers used a least-to-most prompting procedure. This training procedure was used for all five picture stimuli in the study. Participants continued to the test phase following a mastery criterion of 100% correct responding for three consecutive trial blocks.

Test: Bottom-up entailment In this phase, the participants were presented with two picture stimuli (B and C or B and D) that belonged in the same category and were asked, “What do both belong to?” (A). For example, when presented with an airplane (B) and bird (D) and asked, “What do both belong to?” the participant demonstrated the emergence of the mutually entailed response, “HEF” (A). When presented with an airplane (B) and a helicopter (C) and asked, “What do both belong to?” the participant would then respond, “SAN” (A). For these trial blocks, correct and incorrect responses were recorded but no feedback or reinforcement was provided.

Test: Transformation of stimulus function During these trial blocks, participants were presented with all five picture stimuli used in the present study arranged in a randomized array. The participants then engaged in a game of I Spy with the researchers using two arbitrary categories (A) as descriptors for a single picture (B, C, or D). For example, when given the verbal description “I spy with my little eye something that is SAN and HEF” (A), the participant would select the picture of the airplane (B). For these trial blocks, no feedback or reinforcement was provided.

Results and Discussion

The results are shown in Fig. 3. Joseph’s mean correct responding in baseline was 6.25% (range 0%–12.5%) for top-down training trials, 12.5% for bottom-up trials, and 16.7% for transformation trials. In the training phase, Joseph’s mean correct responding for top-down training trials increased to 70% (range 37.5%–100%), and he reached the mastery criteria of 3 trials blocks at 100% correct responding after 10 trial blocks. Following training, probes for bottom-up mutual entailment and transformation of stimulus function were completed. His baseline scores increased for both test probes with mean correct responding at 93.75% (range 87.5%–100%) for bottom-up mutual entailment and 100% for the transformation I Spy task. There is a clear increasing trend from low baseline scores in each phase to high levels of correct responding for all phases following top-down training. William’s mean correct responding in baseline was 4.16% (range 0%–12.5%) for top-down trials, 4.16% (range 0%–12.5%) for bottom-up trials, and 0% for transformation trials. He reached mastery criteria for top-down training within seven trials blocks, with mean correct responding at 83.93%. Following mastery criteria for training, test probes for all three conditions were conducted. William’s mean correct responding was 87.5% (range 75%–100%) for the top-down trials and was 100% for bottom-up mutual entailment and transformation probes. William also displayed a large increase in correct responding between baseline and test phases following top-down training. The third participant, Daniel, had scores of 0% correct responding for top-down and bottom-up entailment trials during baseline phases and a mean of

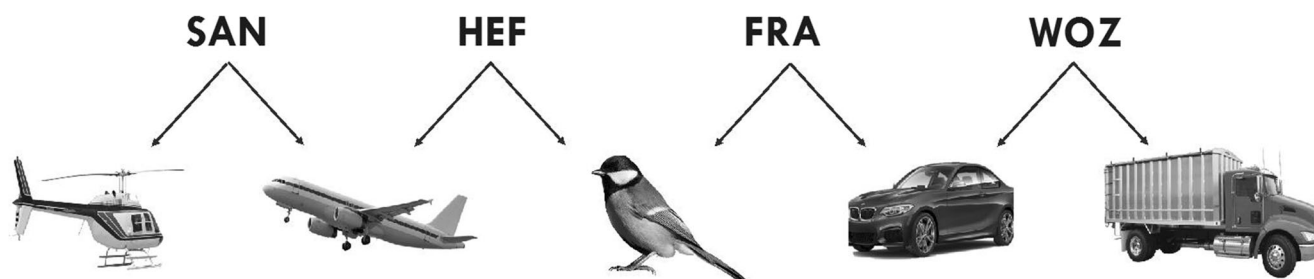


Fig. 2 Stimulus and hierarchical class arrangement used in the present study

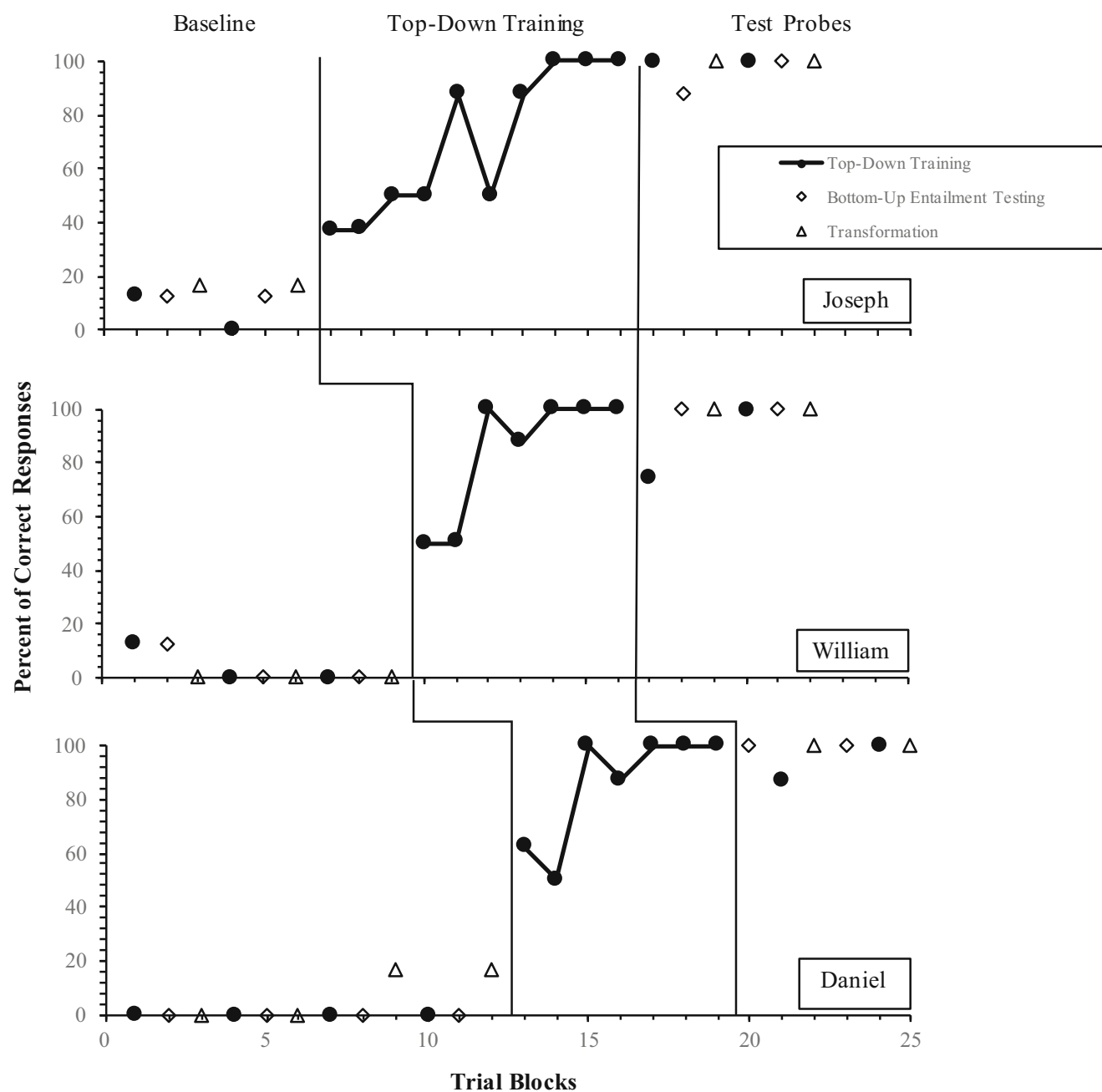


Fig. 3 Percentage correct responding for Joseph, William, and Daniel during baseline, top-down training, and test probe conditions. Closed data points represent trial blocks for trained relations, and open data

points represent trial blocks for untrained relations. Data points with connected data paths indicate training trials, whereas test probes are not connected

8.35% (range 0%–16.7%) correct responding for the transformation I Spy task. Daniel reached mastery criteria for the top-down training phase following seven training trials with a mean of 85.71% (range 50%–100%) correct responding. Following mastery of top-down training trials, Daniel completed test probes for all three conditions, with a mean correct response of 93.75% (range 87.5%–100%) for top-down trials, 100% for bottom-up mutual entailment, and 100% for the transformation task. For all three participants, the percentage

of nonoverlapping data between baseline and training/test conditions was 100%, with all baseline scores below scores during training and test trials. Overall, all three participants in the study were unable to correctly respond to questions in the top-down task, bottom-up entailment task, and transformation I Spy task for the majority of baseline trials, and following top-down training trials, all three participants were able to demonstrate the mutually entailed bottom-up relations and able to accurately identify the correct stimuli during games of I Spy.

Taken together, the results suggest that following training of a hierarchical relation, participants were able to demonstrate the emergence of the mutually entailed relation, as well as the transformation of stimulus function in a game of I Spy, which required multiple stimulus control of arbitrary words. These results suggest that common children's games, a context that is both accessible and socially valid, may provide opportunities to train complex verbal repertoires, extending prior research in this area (Dixon, Blevins, et al., 2019; Dixon, McCord, & Belisle, 2018; Dixon et al., 2016). The current study adds to an RFT analysis of categorization as hierarchical responding using an applied approach, as well as the literature regarding teaching categorization skills to children with autism (e.g., Dixon et al., 2017; Kisamore et al., 2011; Miguel et al., 2008). In addition, the procedures, taken from a standardized curriculum, demonstrate utility for teaching language skills to children with autism, who frequently need training in both higher order language processes and social skills. As part of a comprehensive treatment package, these results provide evidence for the utility of PEAK with children with autism.

Although the present study demonstrates effective procedures, limitations include the use of arbitrary symbols, which restrict its applied utility, and the tests for only the emergence of mutual entailment, although the procedures could be adapted in future research to test combinatorially entailed relations by including a probe of the relation among the lower order stimuli (e.g., "Which two belong together?"). Additionally, as a result of the procedures used in the present study, it is not certain that the mutually entailed relations were the result of hierarchical rather than coordinated responding. However, the participants' performance on the transformation task, selecting one transportation image that was a member of two classes, in the absence of direct reinforcement suggests that responding was hierarchical. Specifically, if the relations were coordinated, then the utterance "I spy with my little eye something that is Ax and Ay" would evoke selection of all stimuli in the array, rather than merely selection of the single stimulus contained in both classes, providing early evidence of unilateral property induction. Future research may address these limitations by using culturally relevant stimuli for children with less sophisticated verbal repertoires who do not have a previous history of playing I Spy, such as the common stimuli "apple," "Elmo," and "red toy truck" as described in the previous example. In addition, by including additional stimuli in a third tier of the hierarchical relational networks—for example, using the stimuli from the present study—a third tier of arbitrary symbols could have been trained as contained by the transportation images. Then, the same test for transformation could be used, except using these additional stimuli, which would have never appeared with the arbitrary words during training. In addition, future research

may examine the effects of this relational training in other novel contexts in addition to the I Spy game—for example, within the game Guess Who. The results of this study suggest that training hierarchical relations to children with autism can result in the emergence of additional untrained hierarchical relations and the transformation of stimulus function to novel, socially relevant tasks.

Compliance with Ethical Standards

Conflict of Interest The last author declares small royalties from the sales of the PEAK curriculum. The remaining authors declare no conflict of interest.

Ethical Approval All procedures performed in studies involving human participants were in accordance with the ethical standards of the institutional and/or national research committee and with the 1964 Helsinki declaration and its later amendments or comparable ethical standards.

Informed Consent Informed consent and assent were obtained from all individual participants included in the study.

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